

Control work 3, and work on the mistakes

Rational equations, systems and inequalities in one paper — then the hour that turns the marks into a plan.

LESSON 38–39

2 × 40 MIN

ALGEBRA 10, NAZORAT ISHI 3

P1 · CHAPTER 1 REVIEW

LESSON CLOCK

3

40'

10'

25'

12'

Instructions	3 min
The paper	40 min
Self-mark	10 min
Rewrite	25 min
Common-error drill	12 min

LEARNING OBJECTIVES

- Apply the rational-equation and rational-inequality methods under time.
- State a domain and reject an extraneous root without being prompted.
- Classify each lost mark as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National	pp. 143–146
Cambridge	Pure Mathematics 1, pp. 24–25
Workbook	Control paper C

01

Explanation

The paper — 25 marks, 40 minutes

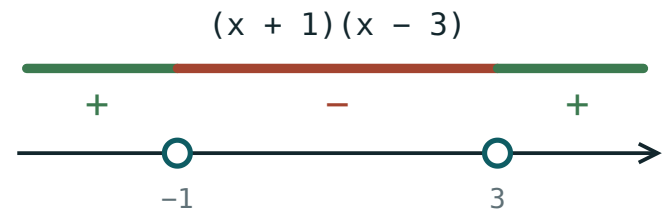
Q	TASK	MARKS	FROM
1	Solve $\frac{3}{x} + \frac{2}{x+1} = 3$, stating the domain	4	L28–29
2	Solve $\frac{x^2}{x-5} = \frac{25}{x-5}$ and justify the rejection	3	L28–29
3	Solve the system $x + y = 8$, $\frac{1}{x} + \frac{1}{y} = \frac{8}{15}$	4	L30–31
4	Solve $\frac{x-4}{x+2} \leq 0$ with a sign chart	4	L32–34
5	Solve $\frac{2}{x-3} > 1$	4	L32–34
6	Solve the system $\frac{x}{x-1} \geq 0$ and $x^2 < 9$	4	L35–37
7	Find the domain of $\frac{\sqrt{x+1}}{x^2-4}$	2	L35–37

WHERE THE METHOD MARKS ARE

Q1 and Q2 each give one mark for the domain line alone. Q4 and Q5 give two marks for the sign chart, whatever the final interval says. Working, not answers.

The four errors this topic produces

ERROR	LOOKS LIKE	KIND
no domain stated	an extraneous root kept	method
cross-multiplying an inequality	$\frac{1}{x} > 1 \Rightarrow 1 > x$	knowledge
denominator zero included	$x \leq 3$ where 3 kills the fraction	method
only one ordered pair	$(3, 5)$ without $(5, 3)$	careless



negative between the roots, positive outside

Questions 4, 5 and 6 in one picture. Two of the marks are for drawing it.

The rewrite

Every question that lost a mark is rewritten in full, with the domain on the first line and the check on the last. Under each rewrite, one sentence beginning “*I lost this because ...*”.

THE DRILL BEFORE THE REWRITE

Ten expressions on the board. For each, the class calls out only **the domain**, in three seconds. Nothing improves this topic faster.

02 Worked examples

EXAMPLE 1 Model answer, Q2: solve $\frac{x^2}{x-5} = \frac{25}{x-5}$.

- ① Domain $x \neq 5$.
One mark for this line.
- ② $x^2 = 25 \Rightarrow x = \pm 5$
- ③ $x = 5$ is excluded by the domain.
- ④ Only $x = -5$ survives.

ANSWER

$$x = -5$$

EXAMPLE 2 Model answer, Q5: solve $\frac{2}{x-3} > 1$.

① $\frac{2}{x-3} - 1 > 0$

Never cross-multiply.

② $\frac{2 - (x-3)}{x-3} = \frac{5-x}{x-3} > 0$

③ Critical 3 and 5, both open.

④ Test 0, 4, 6: $-$, $+$, $-$.

ANSWER

$$3 < x < 5$$

EXAMPLE 3 A learner wrote $\frac{x-4}{x+2} \leq 0 \Rightarrow -2 \leq x \leq 4$. Name the mistake.

① $x = -2$ makes the denominator zero.

② A denominator zero is never included.
Method error.

③ Correct: $-2 < x \leq 4$.

ANSWER

Method error — the excluded endpoint was closed

03 Interactive model

Run the domain drill before the rewrite, not after.

Mark these six answersDomain of $\frac{3}{x} + \frac{2}{x+1}$:

$x \neq 0$

$x \neq -1$

$x \neq 0, -1$

$\mathbb{R}$

 $\frac{x^2}{x-5} = \frac{25}{x-5}$ gives:

$x = \pm 5$

$x = 5$

$x = -5$

no solution

 $\frac{x-4}{x+2} \leq 0$ gives:

$-2 \leq x \leq 4$

$-2 < x \leq 4$

$-2 < x < 4$

$x \leq 4$

2 $x - 3 > 1$ gives:

The system $x + y = 8$, 1 x + 1 y = 8 15 gives:

Domain of $\sqrt{x+1} x^2 - 4$:

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Extraneous root	Chet ildiz	Посторонний корень
Sign chart	Ishoralar jadvali	Таблица знаков
Careless error	E'tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Knowledge gap	Bilim bo'shlig'i	Пробел в знаниях
Mark scheme	Baholash sxemasi	Схема оценивания
Correction	Tuzatish	Исправление

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The domain line is worth:

nothing

a mark on its own

the whole question

half

Cross-multiplying an inequality is:

fine

fine if positive

never allowed with an unknown sign

always wrong

A denominator zero in a $\leq$ answer is:

included

excluded

sometimes included

the answer

A symmetric system is finished when you have:

one pair

both ordered pairs

the sum

the product

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Domain of $\frac{3}{x} + \frac{2}{x+1}$

ANSWER $x \neq 0, -1$

2. Solve $\frac{4}{x} = 2$

ANSWER $x = 2$

3. Solve $\frac{x-4}{x+2} \leq 0$

ANSWER $-2 < x \leq 4$

4. Solve $x^2 < 9$

ANSWER $-3 < x < 3$

5. Solve $x + y = 8, xy = 15$

ANSWER $(3,5), (5,3)$

6. Domain of $\sqrt{x+1}$

ANSWER $x \geq -1$

7. Solve $\frac{1}{x-3} > 0$

ANSWER $x > 3$

Medium · the standard question

7 PROBLEMS

1. Solve $\frac{3}{x} + \frac{2}{x+1} = 3$

ANSWER $x = \frac{3 \pm \sqrt{45}}{6}$ — i.e. $x \approx 1.62, -0.62$

2. Solve $\frac{x^2}{x-5} = \frac{25}{x-5}$

ANSWER $x = -5$

3. Solve $\frac{2}{x-3} > 1$

ANSWER $3 < x < 5$

4. Solve $\frac{x}{x-1} \geq 0$ and $x^2 < 9$

ANSWER $-3 < x \leq 0$ or $1 < x < 3$

5. Domain of $\frac{\sqrt{x+1}}{x^2-4}$

ANSWER $x \geq -1, x \neq 2$

6. Solve $x + y = 8, \frac{1}{x} + \frac{1}{y} = \frac{8}{15}$

ANSWER $(3,5), (5,3)$

7. Solve $\frac{x+3}{x-1} < 2$

ANSWER $x < 1$ or $x > 5$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\frac{1}{x-2} + \frac{1}{x+2} = \frac{4}{x^2-4}$

ANSWER no solution — $x = 2$ rejected

2. Solve $\frac{x^2-9}{x^2-4x+3} \geq 0$

ANSWER $x < 1$ or $x \geq 3$, excluding $x = 3 \rightarrow x < 1$ or $x > 3$

3. Solve $x + y = 10, x^2 + y^2 = 58$

ANSWER $(3,7), (7,3)$

4. Domain of $\sqrt{\frac{x-2}{x+1}}$

ANSWER $x < -1$ or $x \geq 2$

5. Find k so $\frac{x}{x-k} = 2$ has no solution

ANSWER $k = 0$

6. Solve $\frac{1}{x} > \frac{1}{x-1}$

ANSWER $0 < x < 1$ fails; $x < 0$ or $x > 1$

7. Explain why the answer to Q4 is $-2 < x \leq 4$ and not $-2 \leq x \leq 4$

ANSWER The fraction is undefined at -2

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. It is not optional.

1. Rewrite in full every question that lost a mark, with the domain first and the check last.
2. Five problems from the section your knowledge column was heaviest in.

3. Solve $\frac{5}{x+2} \geq 1$ with a sign chart.
4. Solve $x + y = 12$, $xy = 32$, giving both ordered pairs.